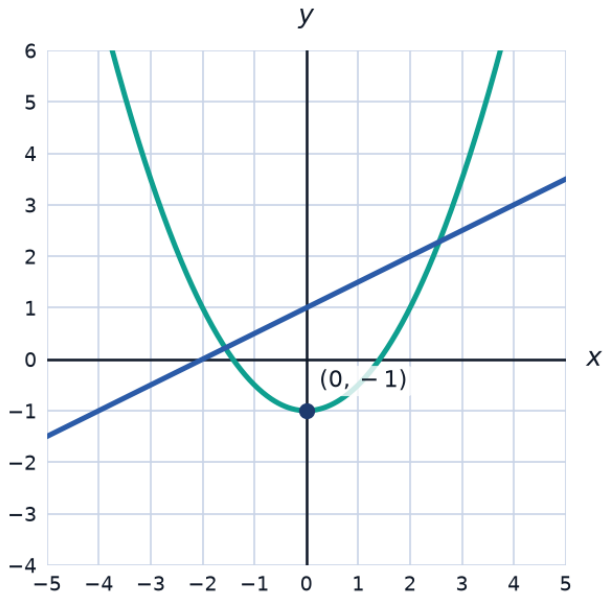
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A car travels 120 miles in 3 hours. Find the average speed.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple of 6 and 8.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Give the equation of the horizontal line through $(0, 5)$.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the distance between $(0, 0)$ and $(3, 4)$.

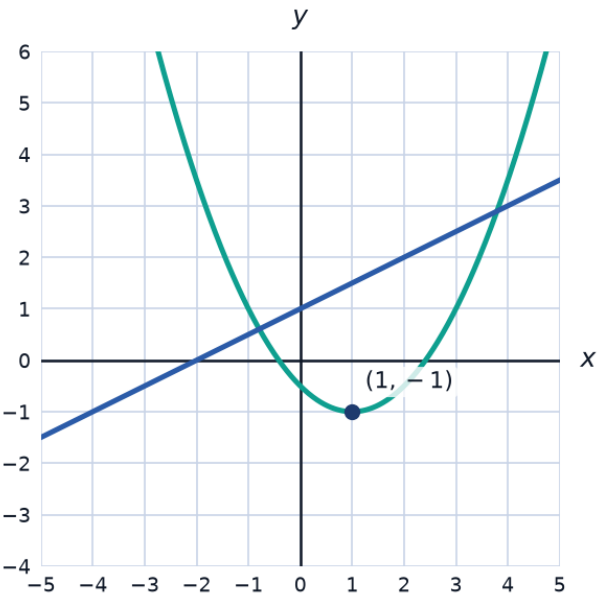
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the interior-angle sum of a pentagon.

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For what value of x is $\frac{1}{x-2}$ undefined?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(2))$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 6 and adjacent 8. Find $\tan\theta$.

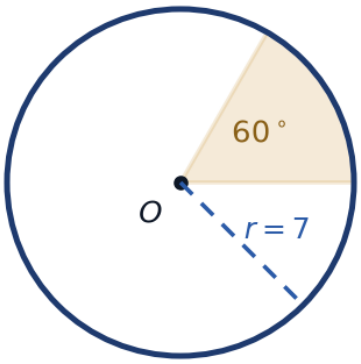
9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the midpoint of $(2, 4)$ and $(6, 8)$.

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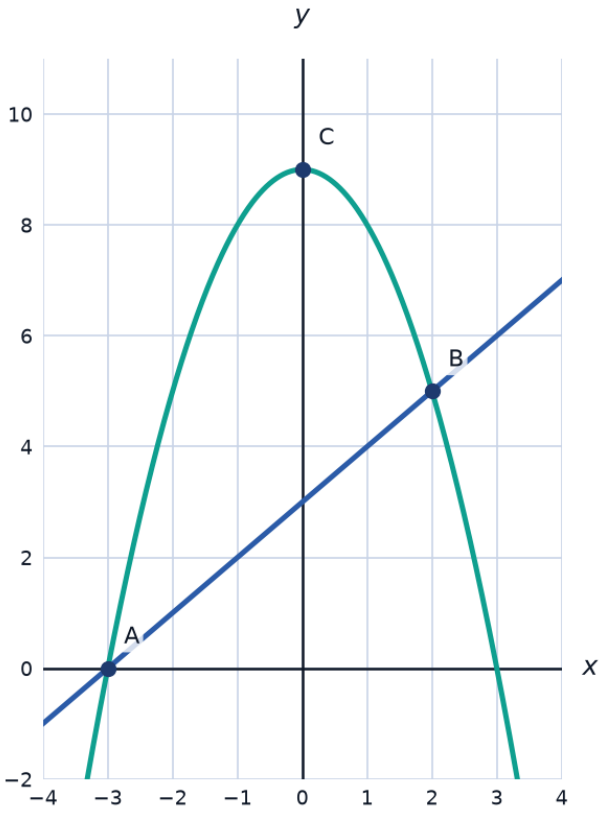
10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the circumference of a circle with $r = 7$. Take $\pi = \frac{22}{7}$.



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

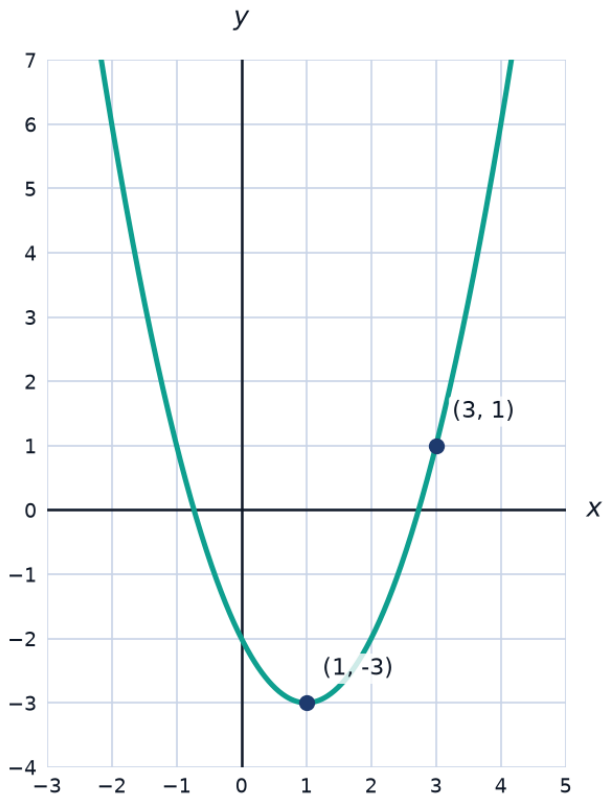
In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of ΔABC? Show your complete algebraic work.



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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

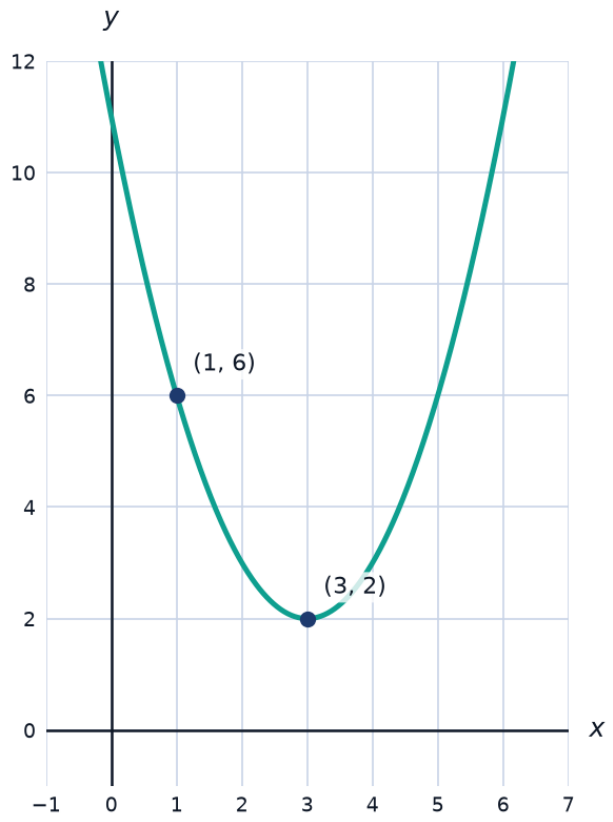
The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with its vertex at $(1, -3)$ and passing through the point $(3, 1)$. What is the value of $f(-2)$? Show your complete algebraic work.



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13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

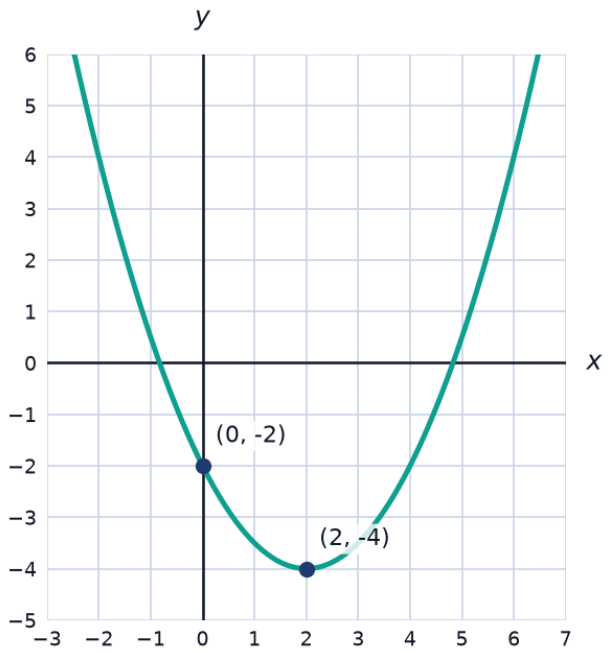
The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with its vertex at $(3, 2)$ and passing through the point $(1, 6)$. What is the value of b? Show your complete algebraic work.



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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -4)$ and y-intercept $(0, -2)$. If the point $(6, c)$ lies on the graph, what is the value of c? Show your complete algebraic work.

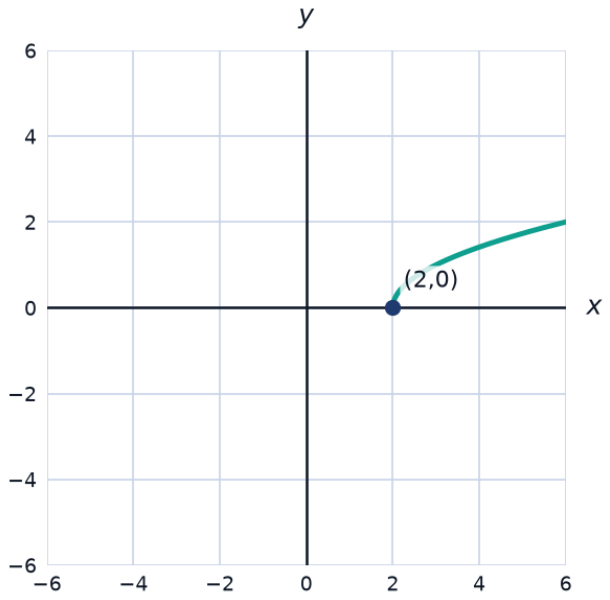


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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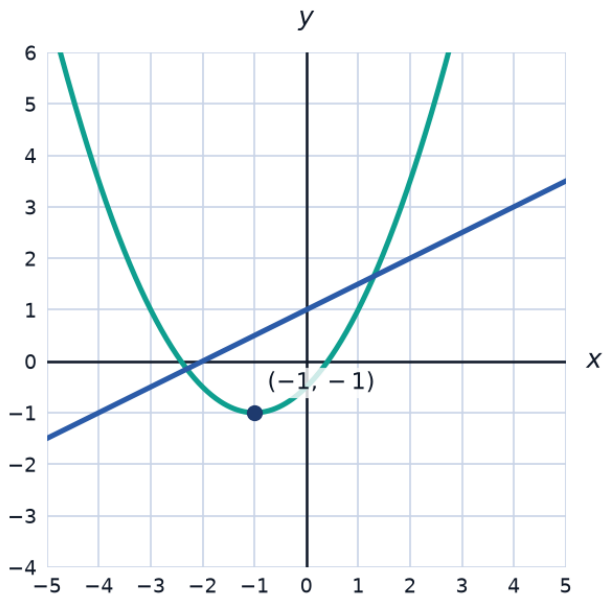
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 2}$? Show your complete algebraic work.



16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

The graph of $y = -f(x)$ is the graph of $y = f(x)$ transformed how? Show your complete algebraic work.



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

The graph of $y = f(-x)$ is the graph of $y = f(x)$ transformed how? Show your complete algebraic work.

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18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

The graph of $y = f(x - 5)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

The graph of $y = f(x - 2)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

The graph of $y = f(x - 1)$ is the graph of $y = f(x)$ shifted how? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $f(x) = 3x^2 - 4x + 7$, what is the value of $f(-2)$?

a. 11

b. 21

c. 27

d. 31

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given the linear function $f(x) = \frac{4x - 7}{3}$, what is the equation of the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x + 7}{4}$

b. $f^{-1}(x) = \frac{3x - 7}{4}$

c. $f^{-1}(x) = \frac{4x + 3}{7}$

d. $f^{-1}(x) = \frac{7x + 3}{4}$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $g(x) = \sqrt{2x + 10}$, what is the domain of $g(x)$ in the set of real numbers?

a. $x \geq -5$

b. $x > -5$

c. $x \geq 0$

d. $x \leq -5$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = 2x + 5$ and $g(x) = x^2 - 3$. What is the value of $(g \circ f)(1)$?

a. 11

b. 46

c. 49

d. 52

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $h(x) = 3x - 1$ and $(h \circ k)(x) = 6x + 8$, which of the following defines $k(x)$?

a. $k(x) = 2x + 3$

b. $k(x) = 2x + 9$

c. $k(x) = 3x + 4$

d. $k(x) = 2x + 2$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the range of the quadratic function $f(x) = -(x - 4)^2 + 9$?

a. $y \leq 9$

b. $y \geq 9$

c. $y \leq 4$

d. All real numbers

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $y = f(x)$ is shifted 3 units to the right and 5 units down. What is the equation of the transformed graph?

a. $y = f(x - 3) - 5$

b. $y = f(x + 3) - 5$

c. $y = f(x - 3) + 5$

d. $y = f(x + 3) + 5$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following functions is an even function, satisfying $f(-x) = f(x)$ for all real x?

a. $f(x) = x^3 - x$

b. $f(x) = 4x^2 + 7$

c. $f(x) = 2x + 1$

d. $f(x) = x^4 - 3x$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the horizontal asymptote of the rational function $f(x) = \frac{6x^2 - 5x + 1}{2x^2 + 3x - 4}$?

a. $y = 0$

b. $y = 3$

c. $y = 2$

d. No horizontal asymptote

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A function is defined piecewise by $f(x) = 2x + 3$ for $x < 1$, and $f(x) = x^2 - 1$ for $x \geq 1$. What is the value of $f(-2) + f(3)$?

a. 7

b. 8

c. 9

d. 11

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $[x]$ denotes the greatest integer function (floor function), what is the value of $[3.8] - [-2.4]$?

a. 5

b. 6

c. 7

d. 1

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c is the piecewise function $f(x) = cx + 4$ for $x \leq 2$, and $f(x) = 3x^2 - 2$ for $x > 2$ continuous at $x = 2$?

a. 2

b. 3

c. 4

d. 5

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = 4f(x)$ is the graph of $y = f(x)$ transformed how?

a. shifted up 4 units

b. vertically stretched by a factor of 4

c. vertically compressed by a factor of 4

d. horizontally stretched by a factor of 4

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) - 2$ is the graph of $y = f(x)$ shifted how?

a. 5 units down

b. 1 units up

c. 2 units up

d. 2 units down

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35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) + 8$ is the graph of $y = f(x)$ shifted how?

a. 8 units up

b. 8 units down

c. 11 units up

d. 7 units down

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) + 6$ is the graph of $y = f(x)$ shifted how?

a. 5 units down

b. 9 units up

c. 6 units down

d. 6 units up

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of $y = f(x) + 2$ is the graph of $y = f(x)$ shifted how?

a. 2 units down

b. 1 units down

c. 5 units up

d. 2 units up

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

The graph of $y = f(x) + 1$ is the graph of $y = f(x)$ shifted how?

a. 1 units down

b. 4 units up

c. 1 units up

d. 5 units down

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

What is the domain restriction for $f(x) = \frac{x+1}{x-9}$? (Give the excluded x-value.)

a. 11

b. 14

c. 3

d. 9

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 11}$?

a. 11

b. 10

c. 26

d. 13

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 24}$?

a. 43

b. 23

c. 7

d. 24

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 26}$?

a. 26

b. 27

c. 12

d. 30

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 28}$?

a. 52

b. 58

c. 7

d. 28

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

What is the smallest value of x in the domain of $f(x) = \sqrt{x - 8}$?

a. 34

b. 14

c. 16

d. 8

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Statistics & Probability

What value of x is excluded from the domain of $f(x) = \frac{1}{x}$?

a. -1

b. 1

c. 5

d. 0

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